MANGALORE INSTITUTE OF TECHNOLOGY & ENGINEERING

(A Unit of Rajalaxmi Education Trust ®, Mangalore)
Autonomous Institute Affiliated to V.T.U, Belagavi, Approved by AICTE, New Delhi
Accredited by NAAC with A+ Grade & ISO 9001:2015 Certified Institution



PROJECT REPORT

23XXSE409 PROJECT PHASE – II

"TITLE OF THE PROJECT IN BLOCK LETTERS"

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

in

[DEPARTMENT NAME]

by

ABC	**4MTxxxxxxxx**
XYZ	**4MTxxxxxxxx**
DEF	**4MTxxxxxxxx**
LMN	**4MTxxxxxxxx**

Under the Guidance of
Dr./ Mr./Ms. OPQ
Professor/Associate Professor/Assistant Professor

DEPARTMENT OF [DEPARTMENT NAME]

(***Accredited by NBA***) **[only if applicable]**

[AY]

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DEPARTMENT OF [DEPARTMENT NAME]

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CERTIFICATE

This is to certify that the Project Phase – II (23XXSE409) work entitled **"TITLE OF THE PROJECT IN BLOCK LETTERS"** is a bonafide work carried out by **STUDENT NAME (4MTxxxxxxx), STUDENT NAME (4MTxxxxxxx), STUDENT NAME (4MTxxxxxxx), STUDENT NAME (4MTxxxxxxx)** in partial fulfillment for the award of degree of **Bachelor of Engineering in [Department Name]** of **Mangalore Institute of Technology & Engineering (Autonomous)** during the year **[AY]**. It is certified that all corrections and suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The project has been approved as it satisfies the academic requirements in respect of project work prescribed for the Bachelor of Engineering degree.

Guide's Name	**HOD's Name**
Project Guide	**Head of Dept. XYZ**

Principal's Name

Principal

DECLARATION

We declare that this Project Phase – II (23XXSE409) report titled "**Title of Project**" submitted in partial fulfillment of the degree of B.E. in **Department Name** is a record of original work carried out by us under the supervision of **Dr./ Mr./Ms. OPQ**, and has not formed the basis for the award of any other degree, in this or any other Institution or University. In keeping with ethical practice in reporting scientific information, due acknowledgements have been made wherever the findings of others have been cited.

Sr. No.	Name	Signature
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ACKNOWLEDGEMENTS

The successful completion of the Project Phase – II (23XXSE409) work titled "Title of Project" would not have been possible without the support of those, whose constant guidance and encouragement helped turn our efforts into success.

We would like to thank our Project Guide **Guide's Name**, Designation, Department Name for his/her constant encouragement and support extended throughout.

We would like to express appreciation to **HOD's Name,** Head of Department of **Department Name** for his/her support and guidance.

We would like to thank our Principal **Principal's Name**, for encouraging us and giving us an opportunity to accomplish the project.

We also thank our management who helped us directly or indirectly in the completion of Phase – II of the project.

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Above all, we extend our sincere gratitude to our parents and friends for their constant encouragement with moral support.

STUDENT NAME (4MTXXXXXXX)

STUDENT NAME (4MTXXXXXXX)

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STUDENT NAME (4MTXXXXXXX)

ABSTRACT

SAMPLE / TEMPLATE NOTE: This is an illustrative abstract. The abstract/synopsis should not exceed 150 words.

Improper segregation of solid waste reduces the effectiveness of recycling and increases the burden on waste-processing facilities. This project presents a computer-vision-based system for classifying waste images into predefined categories using a convolutional neural network (CNN). The system performs image preprocessing, model training, validation and classification through a simple user interface. The project demonstrates how deep learning can support automated waste categorization and provide a basis for smart waste-management applications.

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Abbreviations / Notations

CNN – Convolutional Neural Network

AI – Artificial Intelligence

ML – Machine Learning

ReLU – Rectified Linear Unit

GPU – Graphics Processing Unit

RGB – Red, Green, Blue

API – Application Programming Interface

GUI – Graphical User Interface

CHAPTER 1

INTRODUCTION

SAMPLE / TEMPLATE NOTE: This sample follows the project-report flow. The wording is an example for a CSE related project and should be replaced/adapted for the actual project.

1.1 Background

Solid waste management requires effective segregation of waste into appropriate categories before recycling or disposal. Manual segregation can be time-consuming and may expose workers to unsafe materials. Computer vision and deep learning provide a means of automatically recognizing visual patterns in waste images. CNNs are particularly suitable for image classification because they can learn spatial features from image data.

1.2 Problem Statement

The project addresses the problem of automatically classifying images of common waste items into predefined categories. The system should accept a waste image, process the image, extract relevant visual features using a CNN, and produce the predicted waste category. The intended benefit is a prototype that can support automated segregation and smart waste-management applications.

1.3 Motivation / Need for the Project

Automated classification can reduce dependence on manual visual inspection and can provide consistent classification of images under controlled conditions. A CSE-based implementation also provides an application of machine learning, image processing, software development and model evaluation concepts.

1.4 Objectives

- To develop a CNN-based image classification model for predefined waste categories.
- To preprocess and organize waste-image data for model training and evaluation.
- To train and validate the CNN using suitable optimization and regularization techniques.
- To evaluate classification performance using appropriate metrics.
- To develop a simple interface through which a user can submit an image and obtain a predicted category.

1.5 Scope of the Project

The sample project is limited to image-based classification of predefined waste categories. It focuses on software implementation and model evaluation and does not include physical robotic segregation. The prototype assumes that the input image is sufficiently clear and belongs to one of the categories represented in the training data.

1.6 Significance / Expected Contribution

The project demonstrates an end-to-end CSE solution involving data preparation, deep learning, model training, evaluation and application development. The approach can serve as a foundation for future integration with cameras, IoT devices or automated sorting systems.

1.7 Organization of the Report

Chapter 1 introduces the problem and objectives. Chapter 2 reviews relevant work. Chapter 3 formulates the problem and presents the proposed system. Chapter 4 describes the methodology. Chapter 5 documents implementation and experiments. Chapter 6 presents illustrative results, analysis and discussion. Chapter 7 concludes the report and identifies future scope.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

The literature review examines computer-vision-based waste classification, CNN architectures, image preprocessing and model evaluation. The purpose is to understand existing approaches and identify a suitable direction for the proposed prototype.

2.2 Review of Existing Work

Sl.	Author / Year	Problem / Topic	Approach	Key Finding	Gap / Limitation
1	[Author, Year]	Waste image classification	CNN-based classification	CNN features can support image categorization	Dataset/environment dependent
2	[Author, Year]	Recyclable material recognition	Transfer learning	Pretrained models reduce training effort	Model size may be high
3	[Author, Year]	Automated waste segregation	Computer vision	Visual classification can assist segregation	Real-world variability remains
4	[Author, Year]	Image classification	Data augmentation	Augmentation can improve robustness	Choice of augmentation matters
5	[Author, Year]	Deep learning evaluation	Precision/recall/F1	Multiple metrics provide broader evaluation	Accuracy alone can be misleading

SAMPLE / TEMPLATE NOTE: Replace the placeholder literature entries with actual papers.

2.3 Comparative Analysis

Existing approaches differ in dataset size, number of categories, model architecture, preprocessing and evaluation metrics. For this sample project, a comparatively compact CNN is selected to keep the implementation understandable and suitable for an undergraduate project while still demonstrating the complete deep-learning workflow.

2.4 Research / Design Gap

A project-level implementation should consider not only model accuracy but also preprocessing, training behaviour, generalization, usability and computational requirements. The proposed prototype therefore treats classification as an end-to-end software system rather than only a model-training exercise.

2.5 Summary

The literature review establishes the relevance of CNN-based image classification and identifies the need for a clearly documented, reproducible and evaluable end-to-end implementation.

CHAPTER 3

PROPOSED SYSTEM AND SCOPE

3.1 Problem Formulation

Given an input RGB image I containing a waste object, the system maps I to one of K predefined classes. The classification model learns a function $f(I; \theta) \rightarrow y$, where θ represents learnable parameters and y is the predicted class. The system is evaluated using a held-out test set.

3.2 Proposed Solution

The proposed system consists of five major stages: image acquisition, preprocessing, CNN-based feature learning and classification, performance evaluation, and user-facing prediction. Images are resized and normalized before being supplied to the network. The trained model produces class probabilities through a softmax output layer.

3.3 Requirements

3.3.1 Functional & Non-Functional Requirements

SAMPLE / TEMPLATE NOTE: Replace these illustrative requirements with the actual functional and non-functional requirements of the Project.

The proposed system is defined in terms of both functional and non-functional requirements. Functional requirements specify the operations and services that the system must perform, such as accepting an input image, preprocessing the image, applying the trained CNN model, classifying the image into the appropriate waste category, and displaying the prediction to the user. Non-functional requirements specify the quality attributes and operational characteristics of the system, including performance, usability, reliability, maintainability, scalability, portability, and security. Together, these requirements provide a clear specification of what the system should do and how effectively it should perform its intended functions.

Then the section can proceed with:

Functional Requirements:

FR1 – The system shall allow the user to upload/select a waste image for classification.

FR2 – The system shall preprocess the input image by resizing and normalizing it before classification.

FR3 – The system shall apply the trained CNN model to classify the input image into one of the predefined waste categories.

FR4 – The system shall display the predicted waste category to the user.

FR5 – The system shall provide model-performance evaluation using metrics such as accuracy, precision, recall and F1-score.

FR6 – The system shall allow the trained model to be loaded and used for prediction without retraining.

Non-Functional Requirements:

NFR1 – Performance: The system should provide the prediction within an acceptable response time for a single input image.

NFR2 – Usability: The user interface should be simple and easy to operate without requiring specialized technical knowledge.

NFR3 – Reliability: The system should provide consistent predictions for valid inputs under similar operating conditions.

NFR4 – Maintainability: The software should be modular so that the dataset, CNN architecture or interface can be modified independently.

NFR5 – Scalability: The system should allow additional waste categories and training data to be incorporated with minimal changes.

NFR6 – Portability: The software should be deployable on a standard computer environment supporting the required Python libraries and framework.

NFR7 – Security: User-provided images and application data should be handled appropriately and should not be exposed unnecessarily.

3.3.2 Hardware Requirements

Processor: Intel Core i5 or equivalent

RAM: 8 GB or above

Storage: 20 GB free space

GPU: Optional; NVIDIA GPU recommended for faster training

3.3.3 Software Requirements

Operating System: Windows/Linux

Python 3.x

PyTorch or TensorFlow

NumPy and Pandas

OpenCV / PIL

Matplotlib

3.4 System Architecture / Block Diagram

SAMPLE / TEMPLATE NOTE: Insert the actual architecture figure here. The following text describes what the figure should contain.

Input Image → Preprocessing → CNN Feature Extraction → Fully Connected Classification → Softmax Prediction → Display Predicted Waste Category

3.5 Functional / System Components

- Image input module – accepts an image from the user.
- Preprocessing module – resizes, normalizes and prepares the image.
- CNN module – extracts hierarchical spatial features.
- Classification module – maps learned features to waste categories.
- Evaluation module – calculates classification metrics.
- User interface – displays the predicted category and confidence.

3.6 Scope and Constraints

The sample implementation is intended as an academic prototype. Performance depends on the quality and diversity of the dataset. Images containing multiple overlapping objects, unusual lighting or categories not represented during training may reduce prediction reliability.

CHAPTER 4

METHODOLOGY

4.1 Objective of the Study

The objective is to develop and evaluate a CNN-based image classification system capable of assigning a waste image to one of the predefined classes.

4.2 Method / Approach

- Collect or obtain an appropriate labelled image dataset.
- Inspect class distribution and image quality.
- Resize images to a common input dimension and normalize pixel values.
- Split data into training, validation and test sets.
- Construct a CNN using convolution, ReLU, pooling and fully connected layers.
- Train the model using a suitable loss function and optimizer.
- Monitor validation performance and apply regularization where required.
- Evaluate the final model on unseen test data.
- Integrate the model into a simple prediction interface.

4.3 Measurement / Evaluation Parameters

Metric	Purpose	Typical Interpretation
Accuracy	Overall proportion of correct predictions	Higher value indicates more correct predictions overall
Precision	Correctness of positive predictions for a class	Higher precision means fewer false positives
Recall	Ability to identify instances of a class	Higher recall means fewer false negatives
F1-score	Harmonic mean of precision and recall	Balances precision and recall
Loss	Training objective/error measure	Lower loss generally indicates better fit, subject to generalization

4.4 Tools, Equipment and Resources

The implementation uses Python with a deep-learning framework, image-processing libraries, numerical libraries and visualization tools. The project can be executed on CPU for a small dataset, while GPU acceleration may be used to reduce training time.

4.5 Experimental / Implementation Procedure

1. Prepare labelled image folders according to the selected waste classes.

2. Resize all images to the selected CNN input dimension.

3. Normalize pixel values and apply selected data augmentation only to the training set.

4. Train the CNN for a predetermined number of epochs.

5. Record training and validation loss/accuracy.

6. Select the model based on validation behaviour.

7. Evaluate the selected model on the held-out test set.

8. Generate a confusion matrix and class-wise metrics.

9. Test the prediction interface using unseen sample images.

4.6 CNN Model Architecture

Layer	Example Configuration	Purpose	Output
Input	128 × 128 × 3	Accept RGB image	128×128×3
Conv + ReLU	32 filters, 3×3	Extract local features	Feature maps
Max Pool	2×2	Reduce spatial size	Reduced maps
Conv + ReLU	64 filters, 3×3	Learn higher-level features	Feature maps
Max Pool	2×2	Reduce spatial size	Reduced maps
Fully Connected	128 units	Combine learned features	128
Dropout	0.5	Regularization	128
Softmax	K classes	Class probabilities	K

4.7 Error Sources and Mitigation

- Class imbalance – inspect class distribution and consider appropriate balancing strategies.
- Overfitting – use augmentation, dropout, validation monitoring and suitable model complexity.

- Poor image quality – apply suitable preprocessing and define acceptable input conditions.
- Data leakage – keep test data separate from training and model-selection processes.

4.8 Data Processing and Analysis

The project records accuracy and loss during training and validation. After training, a held-out test set is used to calculate class-wise precision, recall and F1-score. A confusion matrix is used to identify classes that are frequently confused.

CHAPTER 5

IMPLEMENTATION & EXPERIMENTAL WORK

5.1 Implementation Environment

The sample implementation uses Python and a standard deep-learning framework. Images are handled using PIL/OpenCV, numerical operations use NumPy, and plots are generated using Matplotlib. A simple Streamlit interface may be used for demonstration.

5.2 Module-wise Implementation

5.2.1 Data Preparation

Images are organized by class, checked for invalid files, resized and normalized.

5.2.2 Preprocessing

Training images may undergo controlled augmentation such as horizontal flipping or small rotations.

5.2.3 CNN Model

Convolutional layers learn spatial features, pooling reduces dimensionality, and dense layers perform classification.

5.2.4 Training

The model is trained using cross-entropy loss and an optimizer such as Adam.

5.2.5 Evaluation

The trained model is evaluated using unseen test images and multiple classification metrics.

5.2.6 User Interface

The user selects an image and the application displays the predicted category.

5.3 Experimental Setup / Test Cases

For illustration, assume five waste categories: cardboard, glass, metal, paper and plastic. The actual project must report the real number of classes, dataset size and split used.

Parameter	Illustrative Value	Purpose	Replace With Actual

Number of classes	5	Classification categories	Actual value
Image input size	128×128 RGB	Uniform model input	Actual value
Training/Validation/Test	70/15/15	Model development and evaluation	Actual split
Epochs	30	Training duration	Actual value
Batch size	32	Mini-batch training	Actual value

SAMPLE / TEMPLATE NOTE: All numerical experimental settings in this chapter are illustrative placeholders and must be replaced by actual project settings.

5.4 Prototype / System Implementation

The prototype contains an image-upload component, preprocessing routine, trained-model inference routine and result display. The prediction output includes the predicted category and the associated model confidence, subject to the limitations of the trained model.

5.5 Testing and Validation

Test Case	Input	Expected Output	Status (Sample)
TC01	Valid image from known class	Predicted category displayed	Pass
TC02	Image with unsupported format	Input handled or error shown	Pass
TC03	Low-quality image	Prediction displayed with appropriate limitation	Pass
TC04	Unseen test image	Model produces a class prediction	Pass
TC05	Multiple class examples	Predictions correspond to selected classes	Pass

CHAPTER 6

RESULTS, ANALYSIS AND DISCUSSION

SAMPLE / TEMPLATE NOTE: The numbers below are explicitly illustrative sample values to demonstrate how results should be presented. They are not claimed experimental results. All numerical experimental values in are illustrative placeholders and must be replaced by actual project settings/values.

6.1 Results

The trained model should be evaluated on the held-out test set. The following table demonstrates the preferred presentation format using illustrative values.

Metric	Illustrative Value	Purpose	Actual Project Value
Accuracy	92.4%	Overall classification performance	[Enter]
Precision	91.8%	Class prediction correctness	[Enter]
Recall	90.9%	Class detection capability	[Enter]
F1-score	91.3%	Balanced class-wise performance	[Enter]

6.2 Analysis

In the illustrative result, the accuracy is higher than 90%, indicating that most test images are classified correctly. However, accuracy alone is insufficient to establish class-wise performance. Precision, recall and F1-score should therefore be examined for each class, especially when the number of samples differs across categories.

6.3 Discussion

The confusion matrix should be examined to determine which categories are frequently confused. For example, visually similar materials may have overlapping features. Such confusion can be discussed in relation to image quality, lighting, object appearance, dataset diversity and model capacity. These interpretations must be based on the actual results obtained by the project team.

6.4 Comparison with Existing Approaches

Approach	Model Size	Illustrative Accuracy	Observation
Baseline CNN	Small	92.4%	Illustrative baseline
Transfer Learning Model	Medium	94.1%	Illustrative comparison
Traditional Feature +	Small	84.7%	Illustrative comparison

Classifier			

SAMPLE / TEMPLATE NOTE: Replace the comparison with actual experiments or properly cited results from literature. Do not present illustrative numbers as measured results.

6.5 Significance and Implications

A successful prototype can demonstrate the feasibility of using deep learning for image-based waste classification. Practical deployment would require additional testing under varied lighting, camera conditions, object orientations and real-world waste mixtures.

6.6 Potential Applications

- Camera-assisted waste segregation systems.
- Educational demonstrations of computer vision and deep learning.
- IoT-enabled waste bins with image-based classification.
- Decision-support components for waste-management facilities.

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 Summary of Work

The sample project demonstrates an end-to-end CNN-based image-classification workflow. It covers problem formulation, dataset preparation, preprocessing, CNN design, training, validation, testing and user-facing prediction.

7.2 Conclusions

The project illustrates how a CNN can be used to classify waste images into predefined categories. The actual conclusion must be based on the measured results and should state whether the objectives were achieved, together with the important limitations identified during evaluation.

7.3 Contributions

- Developed an academic prototype for image-based waste classification.
- Integrated image preprocessing, CNN inference and performance evaluation.
- Provided a software interface for demonstrating model predictions.

7.4 Utility / Applications

The prototype can serve as a foundation for further research and development in automated waste classification, smart waste management and computer-vision applications.

7.5 Limitations

- Classification is limited to categories represented in the training data.
- Performance may vary under unseen environmental conditions.
- The prototype does not physically sort waste.
- Real-world deployment requires broader validation and appropriate safety considerations.

7.6 Scope for Future Work

- Expand the dataset and include more waste categories.

- Use transfer learning or more advanced architectures where justified.
- Integrate a camera and edge-computing device.
- Connect predictions with an IoT-enabled segregation mechanism.
- Evaluate the system under real-world lighting and object-placement conditions.

References

SAMPLE / TEMPLATE NOTE: Replace the sample placeholders with actual papers/books/web resources used. Number them in order of occurrence and cite them as [1], [2], etc.

[1] [Actual journal/conference paper on CNN-based waste classification].

[2] [Actual paper/book on convolutional neural networks].

[3] [Actual paper on image preprocessing/data augmentation].

[4] [Actual source describing the dataset used].

[5] [Actual documentation/source for the deep-learning framework used].

Appendix:

Include additional materials like circuit diagrams, tables, raw data, pseudocode, etc.

Final Report Formatting Checklist

- A4 size paper (210 × 297 mm), typed on one side.
- Line spacing: 1.5 or double.
- Margins: Left 1.25 inch; Right 1 inch; Top and Bottom 0.75 inch.
- Chapter and section headings follow the VTU-specified font sizes.
- Body text is 12 pt and justified.
- Figures and tables are numbered chapter-wise.
- Only SI units are used where applicable.
- Important equations are numbered in decimal form and equation numbers are right-justified.
- References are numbered in order of occurrence and cited using square brackets.
- Verbatim reproduction is avoided and all sources are properly acknowledged.
- Guide corrections are incorporated before final submission.
- Software project package and README are included as required.
- The final report is hard-bound with plastic covering for departmental/guide submission.